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CHE 313 Transport Phenomena III

Exam #2 (10/30/2025)

Name: Lucia Hernandez

$q = 1$

1. (100 pts) The following saturated liquid feed mixture (Table) is to be separated by a distillation column at 120 psia to obtain 95.0 mol% of pentane (C_5) in the distillate and 96.0 mol% of the hexane (C_6) in the bottom. The column has a partial condenser and a partial reboiler.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	α_{i-ref}	kmol/hr
C3, propane	4.36	5
C4, butane	2.36	15
C5, pentane LK	1.88	25
C6, hexane HK	1.00 (ref)	35
C7, heptane	0.84	20
Total in feed	-	100

- (10 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for mole fraction using stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

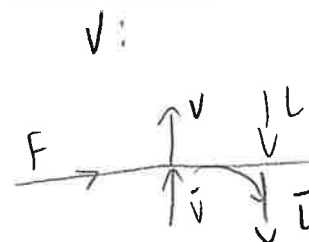
$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where $A=LK$ and $B=HK$, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

- (15 pts - Underwood equation) After drawing mass balance envelope around the feed stage, find ΔV_{feed} . Then, use the following equation to set up equation for ϕ value calculation, so you don't need to calculate it. Define the required all variables and expand the equation from $i = 3 - 7$. Here, $ref = HK$. Four ϕ values (0.8843, 1.354, 2.175, and 4.001) are obtained. Which one is correct?

$$\Delta V_{feed} = \sum_{i=1}^c \frac{\alpha_{i-ref} F z_i}{\alpha_{i-ref} - \phi}$$

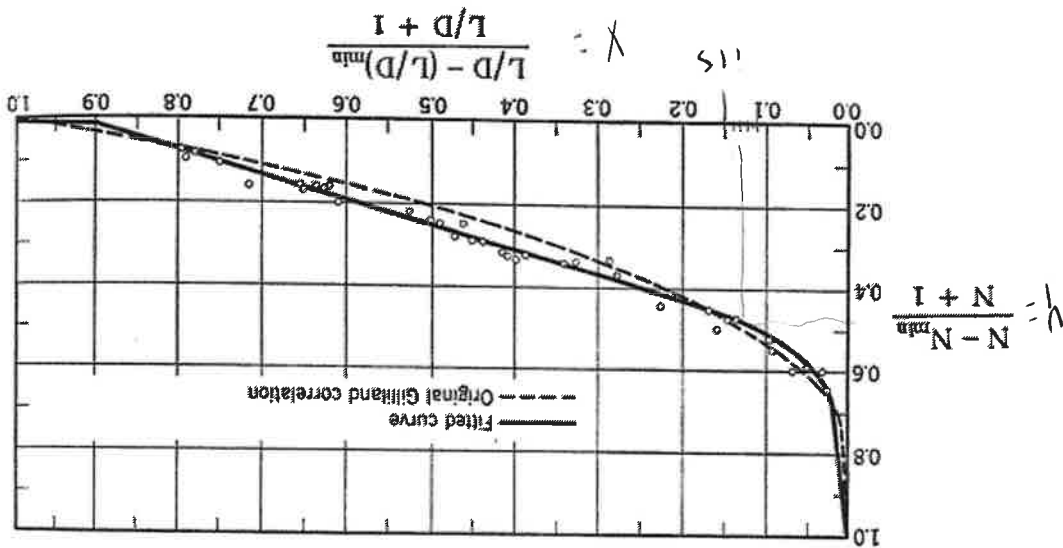
$F(1-q) =$
 $q = 1$
 $0 =$



7) (10 pts- Underwood equation) Find V_{min} and $(L/D)_{min}$. Use mass balance between L_{min} , V_{min} , and D.

$$V_{min} = \sum_{i=1}^c \frac{\alpha_i x_{i,ref} (Dx_{i,dist})}{\alpha_i x_{i,ref} - \phi}$$

8) (10 pts-Gilliland correlation) If $R/R_{min} = 1.3$, find N. Use the following the Fitted correlation curve to calculate N. Use N_{min} and $N_{F,min}$ from (5).



9) (5 pts) Find N_F by using the following ratio equation.

$$\frac{N_{F,min}}{N_F} = \frac{N_{min}}{N}$$

10) (5 pts) Find N_F by using Kirkbride's equation. Compare your result to the value from (9).

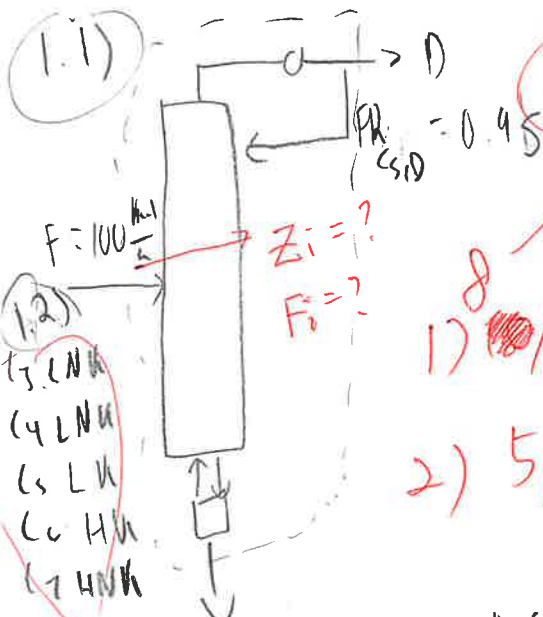
$$\left(\frac{N_F - 1}{N_F} \right) = \left[\frac{B}{D} \left(\frac{z_{LK}}{z_{HK}} \right) \left(\frac{x_{LK,bot}}{x_{HK,dist}} \right) \right]^{0.206}$$

11) (5 pts) Explain whether this distillation process is reasonably economical.

It is reasonably economical since stages needed are not a high number in terms of heavy key and light key.

* $\frac{D}{L}$ low but many stages $\rightarrow$ low cost

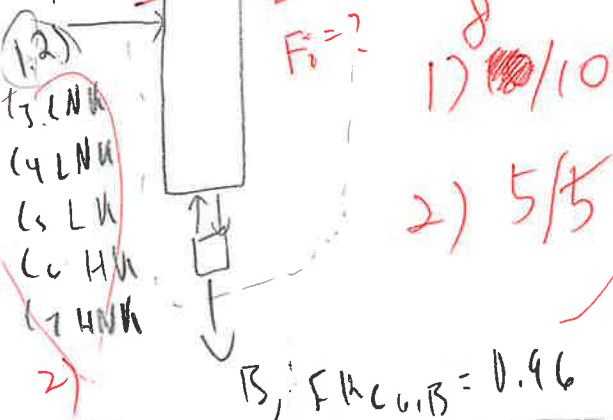
Lucia Hernandez



$$F = D + B$$

$$Fz = Dx_D + Bx_B$$

Assume LNH in distillate only
 $F_{LNH,D} = 1$ $F_{HH,D} = 0$
 Assume LNH in bottom only
 $F_{LNH,B} = 1$ $F_{HH,B} = 0$



Distillate

(1.3) $(3: D \cdot x_{C3,D} = Fz_3 = 5 \frac{\text{kmol}}{\text{h}})$

(4: $D \cdot x_{C4,D} = Fz_4 = 15 \frac{\text{kmol}}{\text{h}}$

(5: $F_{LNH,D} = D \cdot x_{C5,D} \rightarrow D \cdot x_{C5,D} = 0.95(25) = 23.75 \frac{\text{kmol}}{\text{h}}$

(6: $(1 - 0.96) = \frac{D \cdot x_{C6,D}}{Fz_6} \rightarrow D \cdot x_{C6,D} = 0.04(35) = 1.4 \frac{\text{kmol}}{\text{h}}$

(7: $D \cdot x_{C7} = 0$

$D = \sum x_{i,D} = 5 + 15 + 23.75 + 1.4 = 45.15 \frac{\text{kmol}}{\text{h}}$

$x_{C3,D} = \frac{5}{45.15} = 0.11$ $x_{C4,D} = \frac{15}{45.15} = 0.33$

$x_{C5,D} = \frac{23.75}{45.15} = 0.53$ $x_{C6,D} = \frac{1.4}{45.15} = 0.03$

$x_{C7,D} = 0$ $\sum x_{i,D} = 1$

Bottoms:

(3: $B \cdot x_{C3,B} = 0$ (4: $B \cdot x_{C4,B} = 0$

(5: $(1 - 0.95) = \frac{B \cdot x_{C5,B}}{Fz_5} \rightarrow B \cdot x_{C5,B} = 0.05(25) = 1.25 \frac{\text{kmol}}{\text{h}}$

(6: $0.96 = \frac{B \cdot x_{C6,B}}{Fz_6} \rightarrow B \cdot x_{C6,B} = 0.96(35) = 33.6 \frac{\text{kmol}}{\text{h}}$

(7: $B \cdot x_{C7,B} = 20 \frac{\text{kmol}}{\text{h}}$

$x_{C3,B} = x_{C4,B} = 0$

$x_{C5,B} = \frac{1.25}{34.85} = 0.022$

$x_{C6,B} = \frac{33.6}{34.85} = 0.613$

$x_{C7,B} = \frac{20}{34.85} = 0.365$

$\sum x_{i,B} = 1$

$B = \sum B \cdot x_{i,B} = 34.85 \text{ kmol/h}$

$F = D + B$ $100 = 45.15 + 34.85$

$$1.5) N_{min} = \frac{\ln \left[\left(\frac{FR_{LH,D}}{1 - FR_{LH,D}} \right) / \left(\frac{1 - FR_{HH,B}}{FR_{HH,B}} \right) \right]}{\ln \frac{LH - HH}{LS - LC}} = \frac{\ln \left[\left(\frac{0.95}{1 - 0.95} \right) / \left(\frac{1 - 0.96}{0.96} \right) \right]}{\ln 1.88} = 9.7 \approx 10 \text{ stages}$$

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$$N_{F,min} = \frac{\ln \left[\left(\frac{X_{LS,D}}{X_{LC,D}} \right) \times \frac{Z_{LC}}{Z_{LS}} \right]}{\ln 1.88} = \frac{\ln \left[\left(\frac{0.55}{0.03} \right) \times \frac{0.35}{0.25} \right]}{\ln 1.88} = 5.1$$

$$FR_{LS,D} = \frac{D \cdot X_{LS,D}}{F \cdot Z_{LS}} \rightarrow FR_{LS,D} (F \cdot Z_{LS}) = D \cdot X_{LS,D}$$

$$F \cdot Z_{LS} = \frac{D \cdot X_{LS,D}}{FR_{LS,D}} \rightarrow Z_{LS} = \frac{D \cdot X_{LS,D}}{FR_{LS,D} \cdot F}$$

23.75

$$Z_{LS} = 0.25$$

$$FR_{LC,B} = \frac{B \cdot X_{LC,B}}{F \cdot Z_{LC}} \rightarrow FR_{LC,B} (F \cdot Z_{LC}) = B \cdot X_{LC,B}$$

$$F \cdot Z_{LC} = \frac{B \cdot X_{LC,B}}{FR_{LC,B}} \rightarrow Z_{LC} = \frac{B \cdot X_{LC,B}}{FR_{LC,B} \cdot F}$$

133.6

$$Z_{LC} = 0.35$$

or $Z_{LS} = \frac{25}{100} = \frac{F_{LS}}{F}$
 $Z_{LC} = \frac{35}{100} = \frac{F_{LC}}{F}$

6.) $\Delta V_{\text{fired}} = f(1-q) = \sum \frac{\alpha_{i-1} \cdot f_{2i}}{\alpha_{i-1} - \phi}$ Lucia Hernandez
 $q=1 \quad 0 \rightarrow$ Drawing? -2

$$0 = \frac{\alpha_{15-16} \cdot z_{15}}{\alpha_{15-16} - \phi} + \frac{\alpha_{16-17} \cdot z_{16}}{\alpha_{16-16} - \phi} + \frac{\alpha_{17-18} \cdot z_{17}}{\alpha_{17-16} - \phi}$$

$$0 = \frac{1.88 \cdot 0.25}{1.88 - \phi} + \frac{1 \cdot 0.35}{1 - \phi} + \frac{0.84 \cdot 0.7}{0.84 - \phi}$$

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* this one $\downarrow$ ($i=3-7$)

$$0 = \frac{\alpha_{13-16} \cdot z_{13}}{\alpha_{13-16} - \phi} + \frac{\alpha_{14-16} \cdot z_{14}}{\alpha_{14-16} - \phi} + \frac{\alpha_{15-16} \cdot z_{15}}{\alpha_{15-16} - \phi} + \frac{\alpha_{16-16} \cdot z_{16}}{\alpha_{16-16} - \phi} + \frac{\alpha_{17-16} \cdot z_{17}}{\alpha_{17-16} - \phi}$$

($\alpha_{Hh-Hh} < \phi < \alpha_{Lh-Hh}$) ($1 < \phi < 1.88$) $\phi = 1.354$ (correct)

7) use all α for $i=3-7$

$$V_{\min} = \frac{\alpha_{13-16} \cdot D \cdot x_{13,10}}{\alpha_{13-16} - 1.354} + \frac{\alpha_{14-16} \cdot D \cdot x_{14,10}}{\alpha_{14-16} - 1.354} + \frac{\alpha_{15-16} \cdot D \cdot x_{15,10}}{\alpha_{15-16} - 1.354} + \left[\downarrow \right] + \left[\downarrow \right] - 2$$

$$V_{\min} = \frac{1.88 \cdot 23.75}{1.88 - 1.354} + \frac{1 \cdot 1.4}{1 - 1.354} + \frac{0.84 \cdot 0}{0.84 - 1.354} = 80.93 \frac{\text{mol}}{\text{h}}$$

* $L_{\min} = V_{\min} - D = 80.93 - 45.15 = 35.78 \frac{\text{mol}}{\text{h}}$ (method OK!)

78.22

= 0.792
1.073

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$$8) X = \frac{R - R_{min}}{R + 1}$$

$$Y = \frac{1 - X^{0.0021}}{1 - 0.99337 X^{0.0021}}$$

$$N = \frac{N_{min} + Y}{1 - Y}$$

$$X = \frac{\frac{R}{R_{min}} - 1}{\frac{R}{R_{min}} + \frac{1}{R_{min}}} = \frac{1.3 - 1}{1.3 + 1} = 0.15$$

(graph)

$$Y \approx 0.48$$

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Method is ok!

$$N = \frac{N_{min} + Y}{1 - Y} = \frac{9.7 + 0.48}{1 - 0.48} = 19.57 \approx 20 \text{ stages}$$

$$8) \frac{L}{D} = 1.3 \cdot \left(\frac{L}{D}\right)_{min} = 1.3 \cdot 1.73 = 2.25$$

$$① \text{ X-axis: } \frac{R - R_{min}}{R + 1} = \frac{2.25 - 1.73}{2.25 + 1} = 0.16$$

$$\text{Y-axis: } 0.47$$

$$9) \frac{N_{Fmin}}{N_{min}} = \frac{(N_F)}{N} \quad \frac{5.1}{9.7} = \frac{N_F}{19.57}$$

$$N_F = 10.29$$

$$5/5 \quad 9) N_F = N \left(\frac{N_{Fmin}}{N_{min}} \right) = 19.57 \left(\frac{5.1}{9.7} \right) = 10.29$$

$$③ N = \frac{N_{min} + Y}{1 - Y} = \frac{9.7 + 0.47}{1 - 0.47} = 19.18$$

$$10) \frac{N_F - 1}{N - N_F} = \left[\frac{R}{D} \left(\frac{2L_6}{2L_5} \right) \left(\frac{X_{L5,B}}{X_{L6,D}} \right)^2 \right]^{0.206}$$

$$\frac{N_F - 1}{N - N_F} = \left[\frac{54.85}{45.15} \left(\frac{0.35}{0.25} \right) \left(\frac{0.022}{0.03} \right)^2 \right]^{0.206}$$

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$$\frac{N_F - 1}{N - N_F} = 0.98179 \rightarrow N_F - 1 = 0.98179 (19.57 - N_F)$$

$$N_F - 1 = 19.2136 - 0.98179 N_F$$

$$1.98179 N_F = 20.2136$$

$$N_F = 10.20$$

*Results are pretty close in a 10 for N_F .

11) 5/5